

between the piston and the gas is found from  $\oint p dv$ , where  $\oint$  denotes the integration over an engine cycle (720° crank angle for these 4 stroke engines).

#### 4.3.2 Mechanical Efficiency

The **mechanical efficiency** of the engine is, like all efficiencies, a ratio of energy input vs energy output. As the engine operates at a certain (rotational) speed, it is more convenient to use **power** (energy per time) rather than energy for the definition.

Specifically, the mechanical efficiency is defined as the ratio of **brake power** ( $\dot{W}_b$ ) delivered by the engine to the **indicated power** ( $\dot{W}_i$ )—

$$\eta_{mech} = \frac{\dot{W}_b}{\dot{W}_i} \quad (4.7)$$

These terms are defined as follows. Firstly, the work done by the gas on the piston over a single cycle is known as the **indicated work** and is given by —

$$W_i = \oint p dV$$

This is work per cycle but can be converted into **indicated power**,  $\dot{W}_i$ , by multiplying by the number of cycles per second ( $N'$ ):

$$\dot{W}_i = W_i N' \quad (4.8)$$

where for a four-stroke engine with  $n$  cylinders running at  $N$  rpm, in which a cycle is completed in 2 revolutions, we can define  $N'$  as:

$$N' = \frac{N n}{120} \quad (4.9)$$

The **brake power** is the product of the engine torque at the flywheel (or dynamometer) and the angular velocity in rad/s —

$$\dot{W}_b = T \omega \quad (4.10)$$

The difference between indicated work and brake work is the **power dissipated as friction**, that is —

$$\dot{W}_f = \dot{W}_i - \dot{W}_b \quad (4.11)$$